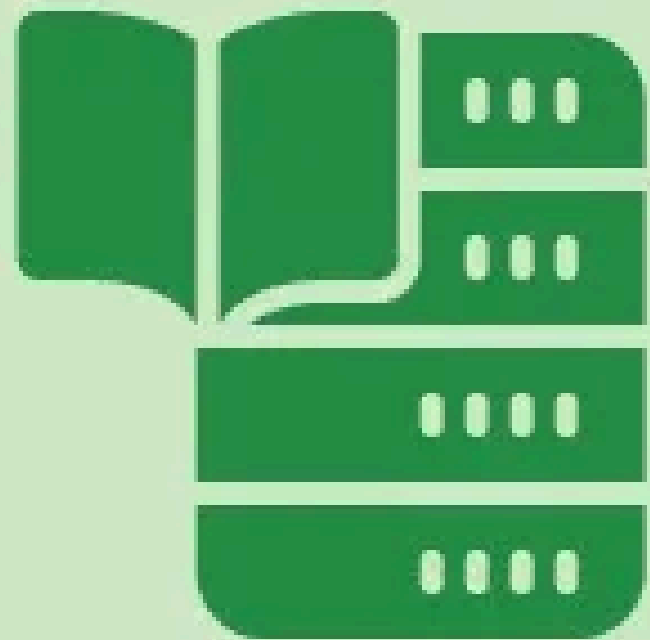


ML ALGORITHMS

Types of Machine Learning



Supervised Learning

Learns from labeled data.



Unsupervised Learning

Finds patterns in data without labels.



Reinforcement Learning

Learns through trial and error.

Machine Learning = Give data → Learn patterns → Make predictions

Algorithm	Simple meaning	Example
Linear Regression	Predict a number	Salary, sales, house price
Logistic Regression	Predict a category	Yes/No, Spam/Not Spam
KNN	Look at nearby examples	Classify a new customer
Naive Bayes	Use probability	Spam detection, sentiment

If a patient checks whether hes going to have heart disease is called
classification(binary)
but whether he could take insurance or not is Regression (how much i
could pay)

Algorithm =
statistics distribution(EDA)
+ mathematical intuition +
dataset workout

Statistical Distribution

Binomial Distribution → coin -> head /tail

Bernouli Distribution → 10 time* Tossing

**Geometric distribution → How many u should toss a coin to get head or
tail**

Linear Regression

- Supervised learning regression algorithm
- ip (x) vs op(y)

relationship between 2 values

- X independent variable
- y dependent variable/
predicted variable/ target
vector

Linear Regression

Purpose: Predicts continuous numerical values (e.g., house prices, temperatures).

How it works: Fits a straight line (best-fit hyperplane) that minimizes the sum of squared differences between predicted and actual values.

Best used for: Continuous target variables with a linear relationship to features. [1, 2, 3]

Input → Model → Number

Example:

Hours studied → Exam score

If students study:

1 hour → 35

2 hours → 45

3 hours → 55

4 hours → 65

5 hours → 75

The model learns a relationship between hours and marks.

Then:

Student studies 5 hours → predict ~80 marks.

That's Linear Regression.

Let me draw the best straight line through these points.

$$y = mx + b$$

Prediction = slope × input + starting point

Symbol	Simple meaning
y	What I want to predict
x	What I already know
m	How much y changes when x changes
b	Starting point

$$y = 10x + 25$$

(if 5 hours)

$$\begin{aligned} y &= 10(5) + 25 \\ &= 50 + 25 \\ &= 75 \end{aligned}$$



Google Colab

 [google.com](https://colab.google.com)

Logistic Regression

Purpose: Performs binary or categorical classification (e.g., spam vs. not spam).

How it works: Uses a logistic or sigmoid function to map linear combinations of input features to probabilities between 0 and 1.

Best used for: Linearly separable classification problems requiring probabilistic outputs. [1, 2]

Linear Regression predicts HOW MUCH. Logistic Regression predicts WHICH CLASS. Hours studied → Pass / Fail

For example:

Student A $\rightarrow$ 0.15

Student B $\rightarrow$ 0.42

Student C $\rightarrow$ 0.78

Student D $\rightarrow$ 0.91

Then we use a threshold.

Usually:

Probability $\geq 0.5 \rightarrow$ Class 1

Probability $< 0.5 \rightarrow$ Class 0

So:

0.78 $\rightarrow$ Pass

0.91 $\rightarrow$ Pass

0.15 $\rightarrow$ Fail

0.42 $\rightarrow$ Fail

DATA



Initial Weight



Make Prediction



Compare with Actual



LOSS



GRADIENT



"Change weight this way"



Update Weight + Bias



Make Prediction



...



FINAL WEIGHTS 

Naive Bayes

Purpose: Handles fast text and categorical classification (e.g., sentiment analysis, document sorting).

How it works: Applies Bayes' Theorem of probability while making a "naive" assumption that all input features are conditionally independent of one another.

Best used for: High-dimensional datasets like text data where feature independence holds reasonably well

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

$$P(B)$$

Probability of A after seeing B = probability of seeing B if A is true × probability of A, divided by probability of seeing B.

Formula	Human language
P(A)	What I believed before
P(B A)	How likely is this evidence if A is true?
P(A B)	What I believe after seeing the evidence
P(B)	How common is this evidence overall?



Google Colab

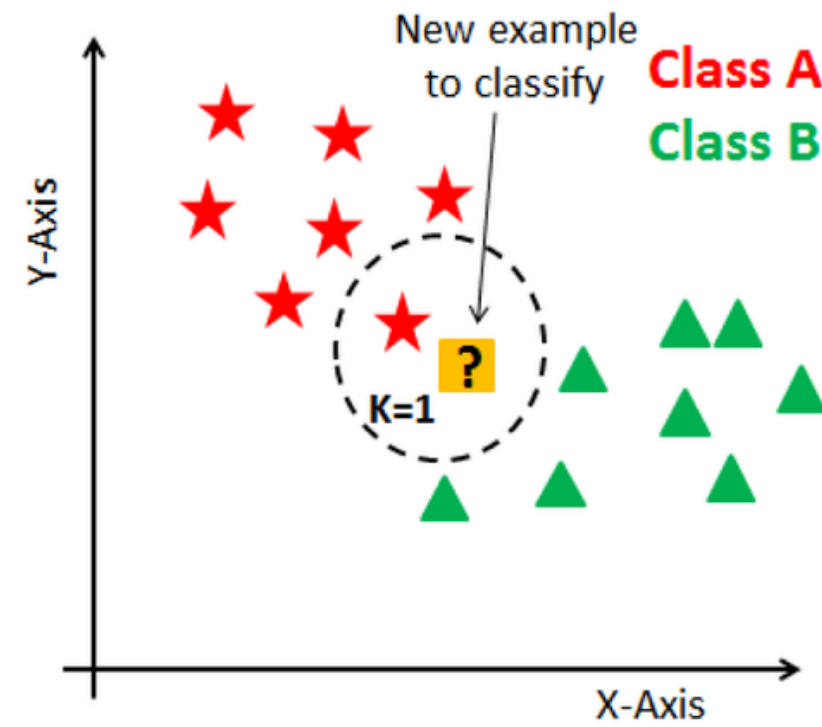
 [google.com](https://colab.google.com)

K-Nearest Neighbors (KNN)

Purpose: Executes non-parametric classification and regression.

How it works: An instance-based "lazy learner" that stores the training set and predicts a label by finding the K closest data points using a distance metric like Euclidean distance.

Best used for: Small to medium, clean datasets with non-linear decision boundaries where no explicit training phase is desired



K-Nearest Neighbor (KNN)



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